

Decentralized Transaction System for Detection and Prevention of Fraud in Crowdfunding Platforms

Bhavana Bafna

Information Technology

Pune Institute of Computer Technology
Pune, India

bhav nabafna.bb@gmail.com

Vedant Daigavane

Information Technology

Pune Institute of Computer Technology
Pune, India

daigavane70@gmail.com

Shlok Shaha

Information Technology

Pune Institute of Computer Technology
Pune, India

shlokshaha1421@gmail.com

Gaurav Shinde

Information Technology

Pune Institute of Computer Technology
Pune, India

gauravshinde1816@gmail.com

Prof. Sachin Shelke

Information Technology

Pune Institute of Computer Technology
Pune, India

Abstract—Crowdfunding via the Internet is a new channel for raising money for startups. It features low barriers to entry, low cost, and high speed, and thus encourages innovation. However, people/entrepreneurs return no value to the contributors after money is raised. This paper focuses on creating a decentralized system using blockchain technologies that will detect and prevent fraud happening in crowdfunding and also be able to provide value back to the contributors either in the form of assets or trust. Blockchain is a decentralized and distributed ledger technology to ensure data security, transparency, and integrity. Because it cannot be tampered with or forged, the technology is deemed to have great potential in the finance industry. This paper examines current problems in the practice of crowd-funding and aims to build a "Decentralized Transaction System" that helps in the detection and prevention of fraud in crowdfunding platforms. The contributors to the campaign will no longer be sending the money directly to the "Idea Person". Instead, they will be sending the money to the Ethereum Contract. This Ethereum contract will have the logic that regulates who can receive the money from the campaign. Here, the "Idea Person" will create the "Spending Request". This would withdraw money from the contract and send it directly to the vendor. Also, already existing contributors have the right to vote (approve) on every "Spending Request". If enough people approve the "Spending Request", the funds will automatically be sent to the vendor. This would eventually allow us to reduce the number of frauds that are being done on crowdfunding platforms.

Index Terms—Crowdfunding, Blockchain, Ledger, Ethereum contract, Decentralized

I. INTRODUCTION

Crowdfunding is the process of funding a project or business venture by raising contributions from many individuals, generally online[1]. Sometimes, people/entrepreneurs return no value to the contributors after money is raised. Also, the money invested by the people on such crowdfunding platforms is used for other purposes rather than using it for registered ideas or startups. There is no way to track the money invested on the crowdfunding platform[2][3]. Only campaign managers can decide how to use the funds invested on the crowdfunding

platform[4]. These kinds of frauds are increasing day by day with the increasing popularity of crowdfunding platforms. Due to this, a large number of people try to avoid contributing through such crowdfunding platforms. Hence, to be able to trust a crowdfunding platform, a platform is required that should have some transaction system that will authorize and authenticate the transactions made for fundraising campaigns so that money donated by the people can only be used for the cause for which they have donated[5].

The existing solution given in Crowdfunding Fraud Prevention using Blockchain [6] just describes the use of blockchains for crowdfunding platforms but does not provide any solid implementation for actually securing the transactions. The paper Secured Crowdfunding Platform Using Blockchain [7] and Crowdfunding Platform Using Ethereum Blockchain Technology [8] describes the idea of using Ethereum blockchain technologies to prevent fraud and secure transactions. This paper describes the solution for detecting and preventing fraud from happening over crowdfunding platforms. The platform uses a decentralized transaction system to secure the transactions happening over the platform, again the transaction to be successfully completed for a spending request, it's necessary to have the approval of a certain no of investors. Thus the transactions made over the platform cannot tamper and funds are used only for the purpose for which it is invested.

II. LITERATURE SURVEY

Now in this related work part, we will discuss some work that has been done in this field.

A. Blockchain-based Crowdlending for Social Businesses

The authors in their paper[1] propose the use of blockchain technology as the foundation for a crowdlending platform to support social businesses. Social businesses are enterprises that aim to solve social or environmental problems while generating profits. However, they often face difficulty accessing

traditional financing options due to their non-profit or social impact focus. The authors in their paper argue that the decentralized nature of blockchain technology can help address trust issues in peer-to-peer lending and reduce transaction costs. The proposed platform would allow individuals to lend money directly to social businesses, cutting out the need for intermediaries such as banks. The authors believe this can help social businesses access the funding they need to grow and have a greater impact. The platform would use smart contracts to automate loan agreements and provide transparency in the lending process. Smart contracts are self-executing contracts with the terms of the agreement between buyer and seller being directly written into lines of code. The authors suggest that using smart contracts can reduce the need for costly intermediaries, provide transparency to all parties involved, and reduce the risk of fraud. The platform would also use blockchain technology to create a decentralized database of all loan transactions. This would help ensure that all loan transactions are recorded and cannot be altered, providing an auditable record of all transactions. Additionally, the platform would allow lenders to track the impact of their investments on the social businesses they fund, giving them greater insight into the social and environmental impact of their investments. The authors in their paper [15] discuss the potential challenges and limitations of using blockchain technology in the context of social business lending. These challenges include regulatory compliance, scalability, and the need for a high level of technical expertise to use the platform. However, the authors suggest that these challenges can be addressed with appropriate measures such as regulation, technological innovation, and user education. Overall, the paper suggests that the use of blockchain technology can help create a more inclusive and sustainable economy by supporting social businesses that have difficulty accessing traditional financing options. The proposed platform can provide a new way for individuals to invest in businesses that align with their values and support social and environmental causes.

B. Blockchain Technology for Financial Inclusion through Crowdfunding

According to the authors in their paper [2] propose that blockchain technology has the potential to improve crowdfunding by promoting financial inclusion. The authors begin by outlining the advantages of crowdfunding and how it can give small and medium-sized businesses (SMEs) and individuals who might not have access to conventional funding sources access to capital. They then draw attention to the drawbacks of conventional crowdfunding platforms, such as their high transaction fees, transparency concerns, and security flaws. The report then discusses the advantages of blockchain technology over conventional crowdfunding platforms. The authors detail how blockchain technology can offer a decentralized crowdfunding platform that is open, safe, and cost-effective. The authors suggest a blockchain-based platform for crowdfunding that makes use of smart contracts to streamline the fundraising process and guarantee accountability and

transparency. To reduce fraud and increase player confidence, the platform has features including identification verification, transaction tracking, and a reputation system. In particular, for underbanked and unbanked people who might not have access to conventional funding sources, the study explores the possible effects of blockchain technology on financial inclusion. According to the authors, platforms for crowdfunding built on blockchain technology can offer these people a more accessible and inclusive type of finance. Ultimately, the article offers a thorough analysis of crowdfunding, its drawbacks, and how blockchain technology might be used to improve crowdfunding and encourage financial inclusion. The authors suggest a useful and efficient method for putting blockchain-based crowdfunding platforms and highlight the potential impact of this technology on financial inclusion.

C. Feasibility of Blockchain for Ethical Project Crowdfunding

The authors of the papers [3] examine the feasibility of using blockchain technology for donation-based crowdfunding of ethical projects. It begins by providing an overview of crowdfunding and blockchain technology, discussing the advantages and limitations of both. The paper then explores the potential benefits of using blockchain for crowdfunding, such as increased transparency, security, and accountability, and discusses the various challenges that need to be addressed, including scalability, regulatory compliance, and user adoption.

The authors of the papers [10] evaluate the feasibility of blockchain-based crowdfunding, the authors conducted a case study of a hypothetical ethical project, examining the costs and benefits of using a blockchain-based platform compared to traditional crowdfunding platforms. The results of the case study suggest that blockchain-based crowdfunding could be a viable option for ethical projects, particularly in terms of transparency and accountability.

The authors of the paper [11] The paper conclude by discussing the implications of these findings for the future of crowdfunding and ethical project funding, highlighting the potential for blockchain technology to revolutionize the way that these activities are conducted. It also identifies areas for further research, including the need for a more robust analysis of the costs and benefits of blockchain-based crowdfunding, and the development of strategies to address the challenges that currently limit its adoption.

D. Live campaign tracking and secure payment processing to improve Crowdfunding Security and Transparency with Blockchain Technology

The authors in papers [4] define crowdsourcing and outline its advantages while also drawing attention to the rising risk of fraud in online crowdfunding projects. There is a need for a more secure and transparent solution because conventional methods of fraud prevention are frequently insufficient. The authors of the paper [12] suggest building a decentralized crowdfunding platform using blockchain technology, specifically Ethereum. The platform makes use of smart contracts

to automatically verify and approve transactions, guaranteeing that only authentic campaign managers are given access to cash. The proposed solution, which consists of a web application and a mobile app for campaign management, is also thoroughly described in the paper, along with its architecture and implementation. The goal of the suggested remedy is to increase stakeholders' trust by bringing transparency, security, and accountability to crowdfunding projects. The authors of the paper [13] give a proof-of-concept implementation and simulation are used by the authors to assess the effectiveness of their suggested method in thwarting fraudulent operations. Overall, the paper offers a novel and workable blockchain-based solution to the issue of fraud in online crowdfunding, and the suggested method has the potential to dramatically increase the accountability and transparency of crowdfunding campaigns.

E. Exploring the Potential of Blockchain in Crowdfunding

The authors of the paper[5] explore the potential use of blockchain technology in crowdfunding, highlighting its potential benefits such as increased transparency, security, and accountability. It examines the current state of crowdfunding and identifies the limitations of traditional crowdfunding platforms. The paper then discusses how blockchain technology can overcome some of these limitations, such as reducing fraud and ensuring that funds are used for their intended purpose. The authors also provide examples of existing blockchain-based crowdfunding platforms and discuss their advantages and limitations. Additionally, the paper explores the legal and regulatory considerations of using blockchain technology for crowdfunding and identifies potential challenges that need to be addressed. Overall, the paper concludes that blockchain technology has the potential to revolutionize crowdfunding by providing a more secure and transparent platform for fundraising.

III. PROPOSED METHODOLOGY

The proposed system architecture contains three major entities Investors (contributors), Campaign managers, and vendors. These entities are interacting with each other, and transactions of the data and digital currency between these entities are secured by using a smart contract (ETH).

Idea Person / Campaign Manager: An idea person is an entity that will register new ideas or startups on the platform. Idea person has the functionality to register new ideas or startups and create spending requests for the new product. A spending request is also called a fundraising request and requires the contributors' validation for further investment. Contributors will be provided with the details of all the Idea persons, if in case a fundraising request fails and some investors still want to invest in the idea / Startup, They can approach the Idea person.

Investors / Contributors: Crowdfunding is the concept where normal people can also act as investors and help the small/medium scale startup raise the funding. This concept was introduced so that the maximum pool of investors can

raise the funds for small/ medium scale startups. Investors can register themselves on the platform and can have a list of registered startups. Each startup has multiple spending requests or fundraising requests. Investors can choose the validated request and raise the fund for that request.

Vendors: The idea of the proposed architecture is to secure the transactions between investors and vendors so that the money donated by investors will reach the correct vendor. Vendors are the end entities that will provide the required resources at the cost of digital currency.

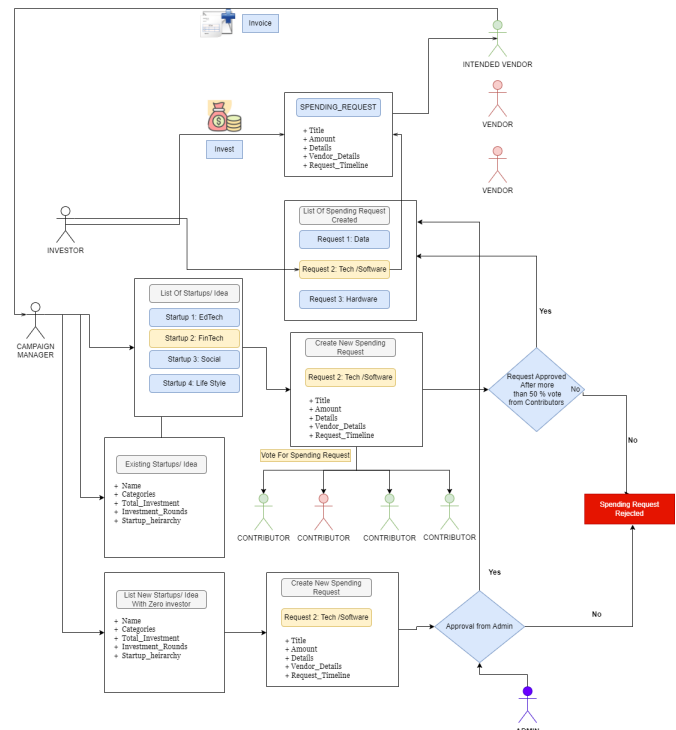


Fig. 1. System Design Diagram

In this project, the user can join as an Investor or as a Campaign Manager or as Vendor. After the joining process, the user has to log in to the platform. After logging in, the user will be directed to the pages based on the user type (Investor, Campaign Manager, Vendor).

When the Campaign Manager logs in, he/she can create a new startup. While creating the new startup, the basic details of the startup are required to be filled in by the Campaign Manager. After the startup is created, it is listed on the startup's page. A Campaign Manager can create multiple startups. These startups can be of multiple domains like EdTech, FinTech, Social, Lifestyle, etc. All these startups will be displayed on the list of startups page.

The Campaign Manager can create a new 'Spending Request' for one of its startups. The 'Spending Request'

consists of a basic funding request that the campaign manager is interested to raise from the investors. It contains the amount to be raised, the reason for raising that amount, and also the Vendor details. The Vendor details are taken so that the amount raised can be directly sent to the Vendor using blockchain.

If the Campaign Manager creates the spending request for the startup with zero investors in it, the spending request will be approved by the Admin team of the platform. After approving the spending request, it will be displayed to the investors to invest in that startup.

When the Investor logs in, he/she will be directed to a page that contains the list of approved spending requests. The investor can invest in the startup with approved spending requests. After investing in the startup, the invested amount will be sent directly to the intended Vendor. After this, the invoice is shared between the Campaign Manager and the Vendor.

The Investors/Contributors can also vote for the spending request created by the Campaign Manager. The request will get approved after more than 50% votes from the contributors/investors. After the request gets approved, it is then listed in the list of spending requests so that investors can invest in the startup. This entire system ensures that the amount invested by the investors reaches only the intended vendor by the use of 'block-chain smart contracts'. This makes the transaction safe and secure as this cannot be tampered with.

Signing the Smart Contract module : Ethereum smart contract called **SpendingRequestContract** is implemented using factory design patterns. Once the request from investors hits the backend API routes, it will have the functionality to create the new transaction using the information submitted from the frontend module. A smart contract is called **SpendingRequestContract** which contains the **createSpendingRequest** methods that will be used to create new contracts and return the deployed contracts respectively. The SpendingRequest contract will contain the description, value, recipient, approved count, voters array, and donators array which will store the information about the spending request. It also contains the vendor address on which the transaction has to be delivered. Once the request is submitted from the frontend module, the contract will get created and stored in the database instance as well as the blockchain network. The current smart contract is deployed on the polygon test net. When transactions are completed at a cost of required gas fees the new block will get added to the previous chain of transactions and the invoice of that transaction is created on the vendor side. The contract contains **getSpendingRequests** method to get the transaction details when it is completed which will be helpful to create the invoice for the transaction.

Backend API and Data persistent module : API gateway is responsible for serving the backend API request from the frontend. Expressjs API module will have the main gateway which will handle the request coming to the main server. The **middleware module** will intercept the request in between the check for the details like Auth token, Campaign manager, Investors, and Vendors private and public key in case of investment workflow and if all the information looks good, it will pass the request to be processed to the backend server. The server will contain multiple routes e.g. **'/campaigns/new'** (to create a new campaign), **'/campaigns/:address'** (Get the campaign details), **'/campaigns/:address/requests'** (Get the spending request for the particular campaigns), etc. **The Data persistence layer** will make a connection to the hosted Atlas server and uses Mongoose ORM to store the data to the MongoDB cluster. There are two connections used in the project **'MONGO_ATLAS_URI'** which makes the connection to the hosted Atlas server and **'MONGO_LOCAL_URI'** which will connect to the locally installed instance of the MongoDB in case of a network connection issue.

Frontend Module : Frontend contains a set of reusable components rendered to provide a view for investors, campaign managers, and vendors registering on the platform. Different routes associated with the backend can be called using the axios library to fetch the data from the backend. The investor will be able to see the investments, campaigns, and approved spending requests on the frontend portal and will have the functionality to invest in the spending request by signing the smart contract. The campaign manager can view the created campaigns and list of investors and their details inside their view. Vendors can see the request for the services made by the campaign manager along with paid spending requests. In that way, vendors will get to know which services they have to provide to the campaign manager.

Existing systems lack the purpose to control where the "Idea Person" can send the money to. Hence this project helps to build a Decentralized Transaction System for the detection and prevention of fraud in crowdfunding platforms that will authorize and authenticate the transactions made for the fundraising campaign so that money donated by the contributors is directly sent to the respective vendors using Smart Contracts.

IV. RESULTS

A. Results of Experiments

The Decentralized transaction system has successfully implemented and deployed a spending request smart contract which will get signed while execution of the transaction. The locally set up application is given to unknown entities Investor, Campaign Manager, and Vendor. The Campaign manager is successfully able to create the account, register a new startup and create spending requests. The Spending request amount and the final vendor address are signed at the time of the

creation of the smart contract. The Investor entity is able to see the spending request on the startup in which he/she has invested and is able to vote on the spending request. The vendor is able to see the money getting transferred to his/her account.

Fig. 2. Create Startup for Campaign Manager

Fig. 3. Startup List for Investor

Fig. 4. Invest by connecting to MetaMask Wallet

B. Result Analysis

ReactJS(NextJS): The existing implementation uses ReactJS (nextJS) as a frontend. The normal implementation of Javascript takes 1215 kb and 1102 ms to load. With React component-based architecture, this loading time is significantly reduced to 405 kb and around 800ms.

Backend API and Database : Implementation uses MongoDB as a database, ThirdWeb SDK is used for integration, and

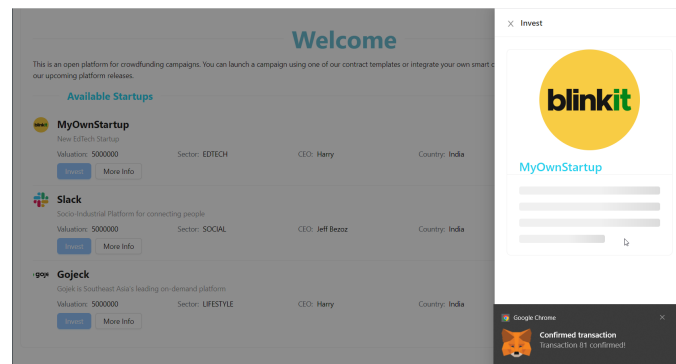


Fig. 5. Successfully Invested by connecting to MetaMask Wallet

Solidity is used for the smart contract. The API serving smart contract is tested against various database and the results for data ingestion is as follows:

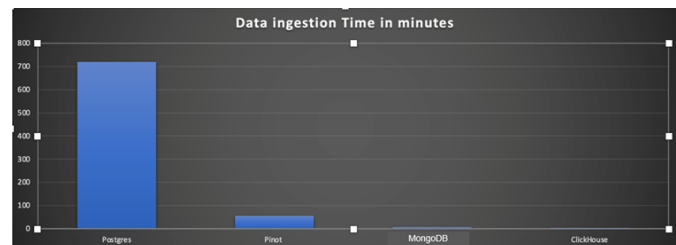


Fig. 6. Data ingestion Diagram

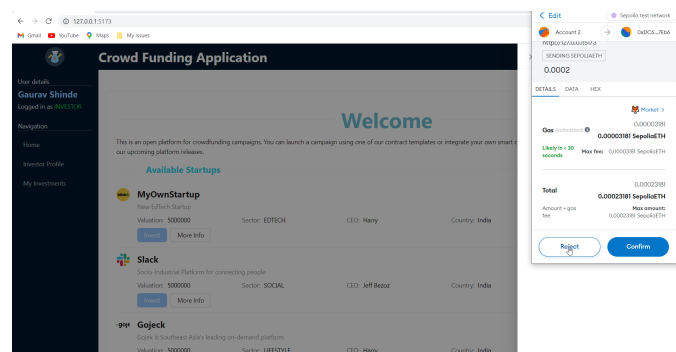


Fig. 7. Database Diagram

CONCLUSION

Reading these research articles helped to dispel many myths about blockchain and its application in crowdfunding, allowing people to better understand what it is. The proposed frame model must be evaluated using simulation technology and era. While being conceptual, the recommended paradigm needs to be thoroughly evaluated before application. It created a decentralized transaction system that aids in fraud detection and prevention on crowdfunding platforms. This transaction system validates and authenticates the fundraising campaign's transactions to ensure that the public's donations can only be applied to the intended good. As a result, we are eventually

```

$ upstart@DAIGAWINELAPTOP:~/BEPROJECT % npm run test
> template@1.0.0 test
> jest

console.log
  logging output: Server is healthy!
    at Object.log (tests/health.test.js:8:13)

console.log
  Campaign manager functionality and validation tested successfully without any failures!
    at Object.log (tests/health.test.js:10:13)

console.log
  Investor validation completed, for investment
    at Object.log (tests/health.test.js:27:13)

console.log
  VOTING-Tested, successfully tested the voting system for investor
    at Object.log (tests/health.test.js:37:13)

console.log
  Admin was able to vote successfully
    at Object.log (tests/health.test.js:46:13)

PASS tests/health.test.js
  Health check on GET /health/check
    ✓ Server is healthy (10 ms)
  Campaign manager check on POST /manager/createCampaign
    ✓ Server is healthy (10 ms)
  Investor investment on POST /investor/invest
    ✓ Server is healthy (10 ms)
  TEST: Investor voting on PUT /investor/vote
    ✓ Server is healthy (11 ms)
  TEST Admin voting system, POST /admin/vote
    ✓ Server is healthy! (11 ms)

Test Suites: 1 passed, 1 total
Tests:       5 passed, 5 total
Snapshots:   0 total
Time:        0.307 s, estimated 1 s
Ran all test suites.
upstart@DAIGAWINELAPTOP:~/BEPROJECT %

```

Fig. 8. Successful passing of Unit Test Cases for backend API

able to lessen the quantity of fraud committed on crowdfunding platforms. As a result, we shall draw the conclusion that blockchain-based completely crowdfunding might be a more complex strategy than previous strategies.

FUTURE WORK

For the detection and prevention of frauds in crowdfunding platforms, we have successfully built a decentralized transaction system that authenticates and authorizes the transactions and eventually helps in the successful prevention of frauds in online crowdfunding platforms. Hence, it helps in reducing the number of frauds that are present in crowdfunding platforms. In the future, we can improve the solution by developing an end-to-end Decentralised App. Additionally, smart contracts can be deployed and tested on the real private Ethereum network. In the future, we can also encourage established investment firms to actively participate in the funding rounds organized on our platform.

REFERENCES

- [1] André Schweizer, Vincent Schlatt, Nils Urbach and Gilbert Fridgen. Unchaining Social organizations – Blockchain as the simple generation of a Crowdlending Platform, 2017
- [2] <https://www.researchgate.net/publication/327586306> Aishath Muneeza; Nur Aishah Arshad; Asma Tajul Arifin, *The Application of Blockchain Technology in Crowdfunding: Towards Financial Inclusion via Technology*, July 2018
- [3] Nida Khan and Rachid Ouaich. Feasibility analysis of Blockchain for Donation-based total Crowdfunding of moral tasks, 2019
- [4] Shivansh Pandey; Shivam Goel; Subodh Bansla; Dhiraj Pandey, Crowdfunding Fraud Prevention using Blockchain, IEEE - 2019
- [5] Bhavya Sri, J. S. Supriya, M. Pranathi Sai, P. Siva Prasad. Crowdfunding the use of Blockchain, 2020
- [6] Utkarsh Tyagi; Prashant Kumar; Raman Singh, Crowdfunding Fraud Prevention using Blockchain, IJETIE August 2020
- [7] Megha Sahu; Ashish Gangaramani; Asha Bharambe, Secured Crowdfunding Platform Using Blockchain, October 2020
- [8] Siddhesh Jadye; Swarup Chattopadhyay; Yash Khodankar; Dr. Nita Patil, Decentralized Crowdfunding Platform Using Ethereum Blockchain Technology, IRJET Apr 2021
- [9] Firmansyah Ashari, Tetuko Catonsukmoro, Wilyu Mahendra horrific, Sfenranto, Gunawan Wang. smart contract and Blockchain for Crowdfunding Platform, 2020.
- [10] Ana Vacca, Andrea Di Sorbo, Corrado A. Visaggio, Gerardo Canfora. a scientific literature review of blockchain and clever settlement improvement, 2021.

- [11] Alexandra Moritz and Joern H. Block. Crowdfunding: A Literature review and studies guidelines, 2016.
- [12] Prof. Sumit S Shevtekar, Saurabh Sahare, Pravin Warghade. Blockchain-based Crowdfunding machine, 2022.
- [13] Susanth kumar reddy kura, Trupthi M. Crowdfunding using Blockchain, 2020.
- [14] Shuai Wang, Liwei Ouyang, Yong Yuan, Xiaochun Ni, Xuan Han, and Fei-Yue Wang. Blockchain-Enabled smart Contracts: architecture, packages and future tendencies, 2019.
- [15] mortgage T.Q. Nguyen a, Thinh G. Hoang b,c, Linh H. Do a, Xuan T. Ngo a, Phuong H.T. Nguyen a, Giang D.L Nguyen b, Giang N.T. Nguyen. The function of blockchain era-based social crowdfunding in advancing social cost advent, 2021.
- [16] Youssef Faqir-Rhazoui, Javier Arroyo1, and Samer Hassan. A comparative evaluation of the platforms for decentralized self-sustaining businesses in the Ethereum blockchain, 2021
- [17] Agrawal, A. K., Catalini, C., and Goldfarb, A. 2014. "Some Simple Economics of Crowdfunding," *Innovation Policy and the Economy* (14:1), pp. 63–97.
- [18] Buterin, Vitalik. 2014b. A next-generation smart contract and decentralized application platform.
- [19] Arunima Ghosh a, Shashank Gupta a, Amit Dua and Neeraj Kumar. protection of Cryptocurrencies in blockchain generation, 2020.
- [20] Liufei Chen, Yushan Li, Hong Wen, WenXin Lei, WenJinHou, Jie Chen. Block Chain Based Secure Scheme For Mobile Communication.
- [21] Kickstarter : <https://www.kickstarter.com/>
- [22] Indiegogo, Inc : <https://www.indiegogo.com/>
- [23] Patreon : <https://www.patreon.com/>
- [24] GoFundMe : <https://www.gofundme.com/>
- [25] RocketHub : <https://www.rockethub.com/crowdfunding>
- [26] Crowdrise : <https://www.crunchbase.com/organization/crowdrise>
- [27] Appbackr : <http://www.appbackr.com/>